CAMPUS YIELDING CONVENIENT LOCKABLE ECO-BIKES(CYCLE)

A PROJECT REPORT

Submitted by

ASWATH P (ATP21EC019)

to

the APJ Abdul Kalam Technological University

in partial fulfillment for the award of the Degree of

BACHELOR OF TECHNOLOGY

in

ELECTRONICS AND COMMUNICATION ENGINEERING

under the supervision of

Ms. GAYATHRI P S.

Department of Electronics and Communication Engineering

AHALIA SCHOOL OF ENGINEERING & TECHNOLOGY

ISO 9001:2015 Certified Institution. Approved by AICTE & Affiliated to A. P. J. Abdul Kalam Technological University

Ahalia Health, Heritage & Knowledge Village, Palakkad - 678557 Ph: 04923-226666 www.ahalia.ac.in

Accredited by NAAC

April 2025

DECLARATION

”I hereby declare that this submission is my own work and that, to the best of my knowledge and belief, it contains no material previously published or written by another person nor material which has been accepted for the award of any other degree or diploma of the university or other institute of higher learning, except where due acknowledgment has been made in the text.”

Place: Palakkad	Signature:
Date:	Name: ASWATH P
	Reg. No.: ATP21EC019

ACKNOWLEDGMENT

First and foremost, we would like to thank the GOD ALMIGHTY for his infinite grace and help without which this project would not have reached its completion.

We would like to express our sincere thanks to the MANAGEMENT of Ahalia School of Engineering and Technology for their support for the completion of this project.

We would like to express our sincere thanks to Dr. P. R. SURESH (Principal, Ahalia School of Engineering and Technology) for the valuable help and support given to us for our project.

We would like to express our sincere thanks to Dr. KRISHNA KUMAR KISHOR (Vice Principal, Ahalia School of Engineering and Technology and Executive Director, Ahalia Group) for the valuable help and support for completion of our project.

We would like to express our sincere thanks to Dr. V. BALAMURUGAN (Head of Department, Electronics and Communication Engineering) for his help and guidance throughout this project.

We would like to express our heartfelt thanks to Ms. GAYATHRI P S., our Project Guide, for her instruction and guidance.

Any work would not be successful if it does not rely on the reference material. In this context, we wish to express our profound sense of gratitude to all teaching and non-teaching staff of our college for giving us a supportive environment in the college.

ABSTRACT

It is challenging for many college students who live away from home to get around between classes, the cafeteria, the campus, the sports arena, and other locations since they lack a vehicle.This project introduces a sustainable bicycle-sharing system designed to revolutionize campus transportation. Utilizing an Android application, the system enables seamless bicycle booking with real-time GPS tracking for user convenience. A smart lock mechanism, powered by an ESP32 microcontroller and solenoid lock, ensures enhanced security and ease of access. The integration of a streamlined payment system further facilitates hassle-free transactions, making eco-friendly commuting both accessible and efficient. By combining smart technology with environmental consciousness, the initiative promotes a greener, healthier, and smarter campus lifestyle..

CONTENTS

LIST OF FIGURES

Chapter 1

INTRODUCTION

In an era where urban mobility solutions are increasingly focused on sustainability and convenience, The ”Campus Yielding Convenient Eco-Bikes” project aims to revolutionize campus transportation by offering a sustainable, eco-friendly alternative to traditional methods of travel. This system leverages technology to provide students, faculty, and staff with easy access to eco-bikes for short-distance travel across the campus. At the core of the project is a user-friendly Android app, built using Kodular, that enables bicycle booking, real-time GPS tracking, and a seamless payment experience.

Key features of CYCLE include:

1. **Smart Locking System**: Secure bike access through a solenoid lock controlled by an ESP32 microcontroller.Lock and unlock functionality directly from the app, ensuring only authorized users can access the bikes.

2. **GPS Tracking**: Real-time location tracking of bikes for users to easily find and access available bikes nearby. Helps in maintaining security and management of the bikes by tracking their usage.

3. **User-Friendly App(Built with kodular)**: Simple and intuitive interface for easy bike booking and management, allows quick rentals, route planning, and payment processing, making cycling accessible to everyone.

4. **Integrated Payment System**: Seamless in-app payment functionality for booking bikes and paying rental charges.Supports secure payment methods, making transactions simple and reliable.

5. **Security Features**: The smart lock ensures bikes are securely locked when not in use, preventing theft. GPS tracking adds an extra layer of security by helping administrators track bikes in case of unauthorized use.

6. **Sustainable and Eco-Friendly transportation**: Promotes green transportation on campus by offering eco-bikes as an alternative to cars and other fuel-based vehicles. Contributes to reducing the carbon footprint and encouraging healthier commuting options.

These features combine to create a comprehensive, efficient, and sustainable bike-sharing solution that enhances mobility on campus while ensuring convenience, security, and eco-friendliness.

1.1 SOCIO-ECONOMIC RELEVANCE OF THE PROJECT

This project has important socioeconomic implications for a number of places. Its implementation may result in significant advantages that go beyond advances in technology, favorably affecting society and economic facets. Key areas where ”bike rental system” contributes are listed below:

- Industry Growth: Eco-Friendly Transportation: As a greener alternative to driving a car, governments and organizations are encouraging cycling, which has resulted in more financing and support for bike rental programs.

- Cost-Effectiveness: Maintenance: Requiring no additional fuels and requiring less maintenance than other cars

- Environmental Protection: Choosing bicycles over cars that emit harmful pollutants helps reduce harmful gas emissions in addition to the previously listed advantages.

1.2 EXISTING SYSTEM

In the current market, there are various cycle rental system that offer advanced features.Key aspects of existing cycle rental system:

- Features: Offers both traditional bicycle and electric options, with various pricingplans, including day passes and annual memberships.
- Drawbacks: Geographic Restrictions: The service area is concentrated in certain neighborhoods, leaving some parts of the city underserved, especially in outer boroughs.

1.3 MOTIVATION

The motivation behind the bicycle rental system stems from a critical need to enhance current exploration and navigation capabilities across various sectors. This need is driven by several factors:

1. **Sustainability**
    - Environmental Impact: Encourages eco-friendly transportation, reducing carbon footprints and traffic congestion.
    - Urban Mobility: Provides a green alternative to cars, contributing to cleaner air and healthier cities.
2. **Health Benefits**
    - Promotes Physical Activity: Encourages users to engage in cycling, improving overall health and fitness.
    - Mental Well-Being: Cycling can reduce stress and enhance mood, providing a mental health boost.

3. **Economic Opportunities**

    - Local Business Support: Can drive traffic to local shops and restaurants, boosting the local economy.
    - Job Creation: Opportunities for maintenance, customer service, and management roles within the rental system.

4. **Convenience and Accessibility**

    - Flexible Transportation: Offers an easy way to navigate urban areas, especially in congested cities.
    - Last-Mile Solution: Complements public transportation by solving the “last mile” challenge, making it easier for people to reach their destinations.

5. **Community Engagement**

    - Building Community: Encourages social interaction and fosters a sense of community among users.
    - Events and Activities: Can facilitate community events like group rides, workshops, or educational programs about cycling safety and maintenance.

Chapter 2

LITERATURE SURVEY

[5]Bicycle Renting Service or sharing service is about renting a cycle. Bicycle sharing schemes offer a minimal effort and ecologically helpful mean of transportation for short journeys. It can likewise be utilized as a shared mode to other open travel, for example, transports, neighborhood trains. Bicycle sharing systems unite the benefits of open and private transportation to all the more likely adventure the given transportation foundation. This paper high point smart bicycle sharing service which is totally accessed by android application. This research paper also states facts and figures that bicycle sharing services are environmentally friendly and also great for an individual health. The main scope of this bicycle renting schemes is to implement it to reduce environmental pollution and for health purposes. This scheme is smart bicycle renting scheme in which GPS tracking, QR scanning, online payment, automatic locking/unlocking through android application, and all features on just one android application. This scheme also has admin website where all the information of user and bikes are stored also history of bikes, payment and rides for security purpose. From a distinctive individual's perspective, bicycle share systems take out the burden of bicycle proprietorship, the need to look for parking places, and the dread of burglary. It helps in decreasing rush hour jam clog as number of vehicles on street can be fundamentally reduced.

[9]With the rising concerns about urban congestion, pollution, and the need for sustainable transportation, cycle-sharing systems have emerged as a popular and eco-friendly solution in urban areas. The GlideX proposed in this paper addresses these challenges by harnessing the power of IoT and integrating advanced sensor technologies to provide a smart and efficient cycle rental service. The core of GlideX is a development board, a versatile and powerful platform that enables seamless communication between the cycles and a central server. This intelligent bicycle is designed by integrating various sensors for real-time monitoring of the cycle's orientation and movement, the actuator for the locking system, providing a reliable and efficient way to secure the cycle when not in use and for detecting the status of the cycle's odometer. This smart bicycle creates a comprehensive and intelligent system that not only facilitates smooth rentals but also enhances the overall safety and reliability of the cycle-rental service. The GlideX bicycle rental system represents a significant step towards revolutionizing urban mobility. The implemented Glidex is tested to verify the various features and screenshots of the results are presented. By prioritizing sustainability, technology, integration and community engagement, this initiative will transform the way people commute in urban areas, promoting a greener and healthier future.

[1]This paper proposes a knowledge-based model that can be used digitally via a smartphone application to service the vehicle rental system in Malaysia called EZGO. EZGO is a website that allows consumers to look for vehicles such as cars, bikes and rental vans that can have the most satisfactory outcome and avoid the rejection of unavailable car rental by providing substitute vehicles that are close to the needs of the customer. Our research reveals the analysis and comparison of the problems faced in the general car rental system for the search mechanism and identifies the value of the rental car. The project implemented an agile approach for the design and development of mobile apps, developed UML diagrams for the car rental system, and performed a survey of prospective customers using questionnaires. This smartphone application is particularly relevant in Malaysia as it offers an alternative option for Malaysian drivers and drivers to provide personal transportation without the need to purchase and own for themselves.

[10]In this paper, we proposes a crowdsourced bicycle monitoring and finding (CBMF) framework with riding user recognition for rider safety and bike security based on Internet of Things technologies. According to our review of relevant research, CBMF is the first framework that provides the following features: i) it can use the inertial data to determine the normal/dangerous state of the bicycle, ii) it can recognize whether the rider is the owner based on the inertial data of cycling, and iii) if the bicycle equipped with the IoT device is stolen/missing, the current locations of the stolen bicycle can be reported by surrounding smartphones that receive the iBeacon signals with an abnormal bit proactively broadcasted by the equipped IoT device. In particular, the densely connected convolutional network is explored in CBMF to extract diverse features of riding for improving rider recognition accuracy. An Android-based prototype with the Jetson Nano developing board is implemented to verify the feasibility and superiority of our framework. Experimental results show that our developed deep learning model outperforms existing methods and can accurately recognize the riding user (with over 90precisely locate the position of the bicycle (reducing about 50which increases the possibility of getting back the stolen/missing bicycle.

[7]In recent years, Internet-supported non-dock bicycle-sharing service had emerged and gradually become warmly applauded by the public. In the non-dock bicycle sharing system planning, the definition of coverage area and the division of sub service areas are highly essential and fundamental, which get little attention in the existing studies. In this paper, we propose a market-oriented methodology for non dock bicycle-sharing system planning. Firstly, the potential demand of bicycle-sharing use is identified using mobile phone data. Then, a link network is constructed to represent the spatial distribution of potential demand. The algorithm of community detection is applied to the link network to discover the densely connected nodes in the area and define the division of sub service areas. Finally, the potential patterns of bicycle-sharing use, the spatial distribution of the potential demand, and the potential emission reduction are analysed based on the resulting division of sub service areas. The proposed method is tested using a data set of approximately 34 million GPS trajectories obtained in Tokyo. The area of Tokyo is subdivided into 21 sub service areas. Suggestions for bicycle management, infrastructure development, and the bicycle-sharing system planning are given regarding sub service areas with different properties. The potential emission reduction for the introduction of bicycle-sharing system is calculated in the area of Tokyo.

Chapter 3

PROJECT DESCRIPTION

3.1 WORKING PRINCIPLE

1. **Open App**

    - **User Action:** The user opens the bike-sharing app on their smartphone.
    - **GPS Activation:** The app automatically activates the phone's GPS to determine the user's current location.
    - **Map Display:** A map interface displays nearby bikes, highlighting their locations with icons. Users can also see their current location marked on the map.

2. **Find Bike**

    - **Locate Idle Bike:**The user browses the map to identify available bikes nearby. Each bike icon may show additional information, such as battery status or distance.
    - **Navigation:** Users can tap on a bike icon for directions or an estimated walking time to reach the bike.

3. **Establishing Bluetooth Connection**

    - **Arrival at Bike:**Once the user arrives at the bike, they locate unique bluetooth id affixed to the bike frame.

- **Connecting Process:** The user should establish a connection with the cycle using bluetooth. Each cycle is given a unique id using which the user can establish connection within the app.
- **Successful Connection:** Once the connection is established the user can unlock the cycle and star the ride
- **Unlock Confirmation:** The bike's lock disengages, and the app displays a message confirming that the ride has started.
- **Failed Scan:** If the scan is unsuccessful, an error message appears, prompting the user to try again.

4. **Start Ride**

- **Ride Initiation:** Once the bike is unlocked, the user can start riding.
- **Charging Begins:** The app begins tracking the ride duration and automatically starts charging the user based on the rental rates, which could be a per-minute or fixed fee.
- **Real-Time Updates:** The app may provide real-time notifications about ride stats, such as time elapsed and estimated cost.

5. **End Ride**

- **Completing the Ride:** When the user finishes their ride, they can end it using the app.
- **Automatic Locking:** The bike's lock engages automatically once the ride is ended.
- **Cost and Duration Summary:** The app displays a summary of the ride,including: o Total cost o Duration of the ride o Distance traveled (if applicable)

- **Payment Process:**The user can confirm payment through the app, which may save payment details for future rides.Which can help the user to raise complaints, if any in the future.

3.1.1 Block Diagram

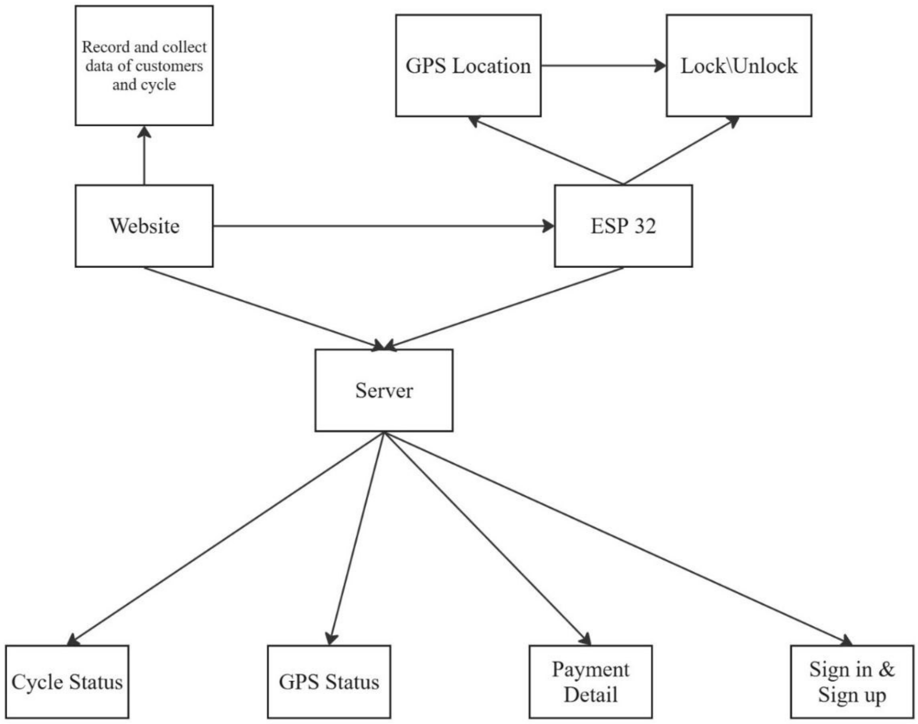

Figure 3.1: Block Diagram

3.1.2 Flow Chart

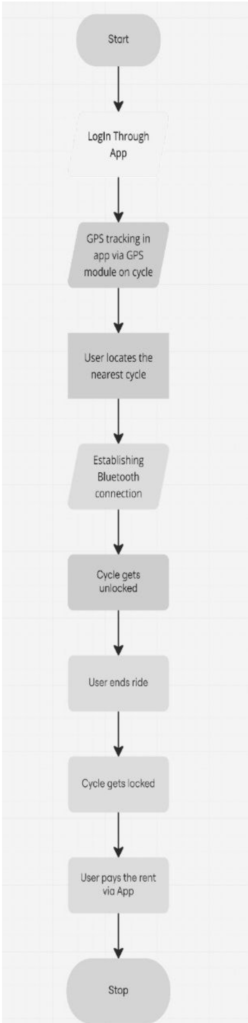

Figure 3.2: Flow Chart

3.2 SOFTWARE REQUIREMENTS

- Kodular
- Firebase

3.3 SOFTWARE DESCRIPTION

3.3.1 Kodular

Kodular is a free online suite for mobile apps development. It mainly provides an online drag-and-drop Android app creator, on which everyone can create any kind of app without programming a single line of code.Some of its key features include:

- Block-based code system: You can join blocks to program your app. Drag and drop: You can customize apps by dragging and dropping blocks.
- **Screens:** You can switch between different screens in your app.
- **Designer:** You can design the user interface (UI). Event-driven approach: The app's behavior is executed based on events that occur, such as the user clicking a button.
- **Material Design:** All apps made in Kodular have a Material Design interface.
- **Compatibility:** Kodular supports all Android phones and tablets running Android 4.4 KitKat or later.
- **Monetization options:** You can add components to your app to benefit from it, such as in-app purchases or ads.
- **Push notifications:** You can use the Push Notification component powered by OneSignal.
- **Community:**You can get help, submit ideas, and report bugs.
- **IDE:** You can create and distribute your own extensions.
- **Store:** You can publish your apps to a specific store.

3.3.2 Firebase

Firebase is a comprehensive app development platform provided by Google that helps developers build, improve, and grow applications. It offers a suite of tools and services designed to streamline the development process and enhance the user experience. Here are the key features and components of Firebase:

- Real-Time Database: Firebase provides a NoSQL cloud database that enables real-time data synchronization between clients and servers. This allows developers to store and sync data across all clients in real-time, making it ideal for applications that require instant updates, such as chat applications or collaborative tools.
- Cloud Firestore: It is a flexible, scalable database for mobile, web, and server development. It allows for structured data storage and supports advanced querying capabilities. Firestore is optimized for performance and can handle complex data structures, making it suitable for a wide range of applications.
- Authentication: Firebase Authentication simplifies the process of implementing user authentication in applications. It supports various authentication methods, including email/password, phone numbers, and social media logins (such as Google, Facebook, and Twitter). This service provides a secure way to manage user identities.

3.4 HARDWARE REQUIREMENTS

- ESP32
- GPS neo 6M
- Solenoid lock
- DC motor

3.5 HARDWARE DESCRIPTION

3.5.1 ESP32

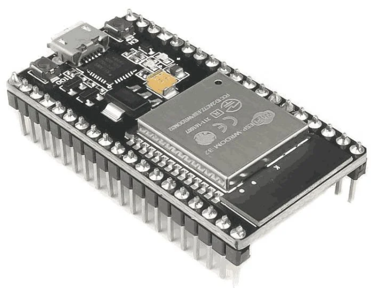

Figure 3.3: ESP32

The ESP32 is a low-cost, low-power system-on-chip (SoC) microcontroller with integrated Wi-Fi and Bluetooth connectivity, designed for various IoT, mobile, and wearable applications, offering features like dual-core processing, numerous GPIO pins, and support for various interfaces. It features a dual-core Tensilica Xtensa LX6 microprocessor, offering powerful processing capabilities for complex tasks

- Processors: CPU: Xtensa dual-core (or single-core) 32-bit LX6 microprocessor, operating at 160 or 240 MHz and performing at up to 600 DMIPS
- Memory: 520 KiB RAM, 448 KiB ROM
- Wireless connectivity: Wi-Fi: 802.11 b/g/n and Bluetooth: v4.2 BR/EDR and BLE (shares the radio with Wi-Fi)

3.5.2 GPS Neo6M

The NEO-6M GPS Module with EPROM is a compact and versatile Global Positioning System (GPS) module that integrates a GPS receiver and non-volatile memory (EPROM) for storing configuration settings or other data.

- Receiver type: 50 channels, GPS L1(1575.42Mhz) C/A code, SBAS:WAAS/EGNOS/MSAS

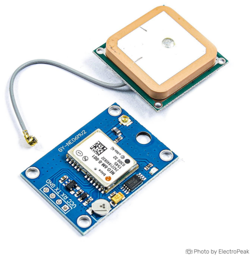

Figure 3.4: GPS Neo6M

- Operating temperature: -40°C 85°C
- Operating voltage: 2.7V 5.0V(power supply input via VCC)
- Operating current: 45mA
- TXD/RXD impedance: 510Ohms

3.5.3 Solenoid Lock

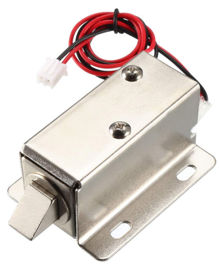

Figure 3.5: Solenoid Lock

Solenoid Lock is an electromechanical locking device designed for secure access control. When activated, it uses an electric current to generate a magnetic field, allowing the lock's bolt to be engaged or disengaged. This versatile lock provides a reliable and controlled locking mechanism suitable for various applications, from doors to cabinets. Its efficient operation, combined with its robust construction, makes the Solenoid Lock a fundamental component in ensuring security and controlled access in both residential and commercial environments.

- Operation Voltage:12VDC ±10 (you can use 9-14 DC volts, but lower voltage results in weaker/slower operation)
- Current: 500mA
- Power: approx. 6W
- Working Temperature:-25ºC +80ºC

Chapter 4

RESULTS AND DISCUSSIONS

This project involves the development of a user-friendly android application for renting bicycles, designed using kodular. The platform aims to facilitate easy access to bicycle rentals while providing users with essential information about cycling routes and destinations within a campus.The result of the "Campus Yielding Convenient Eco-Bikes (CYCLE)" project, which incorporates an Android app, payment system, GPS tracking, login/signup, and a smart lock mechanism controlled by ESP32 with a solenoid lock via Bluetooth, would be a functional and user-friendly system for managing eco-friendly bike rentals on a campus or designated area Key Features:

1. **Android App Development (Using Kodular):**

   - User Interface (UI): The app will have a clean and intuitive interface designed in Kodular, which allows for easy navigation and access to various features like bike rental, payment, login/signup, and tracking.
   - Login and Signup: Users can securely register and log into the system to access bike rentals. This would likely include storing user credentials and possibly integrating with Firebase for authentication and user management.
   - Payment System Integration: The app would include a seamless payment gateway integration (like Google Pay, Stripe, or PayPal), allowing users to pay for their bike rentals directly through the app.
   - GPS Tracking: GPS functionality would allow users to track available bikes and determine the nearest bike location. This feature would also be helpful

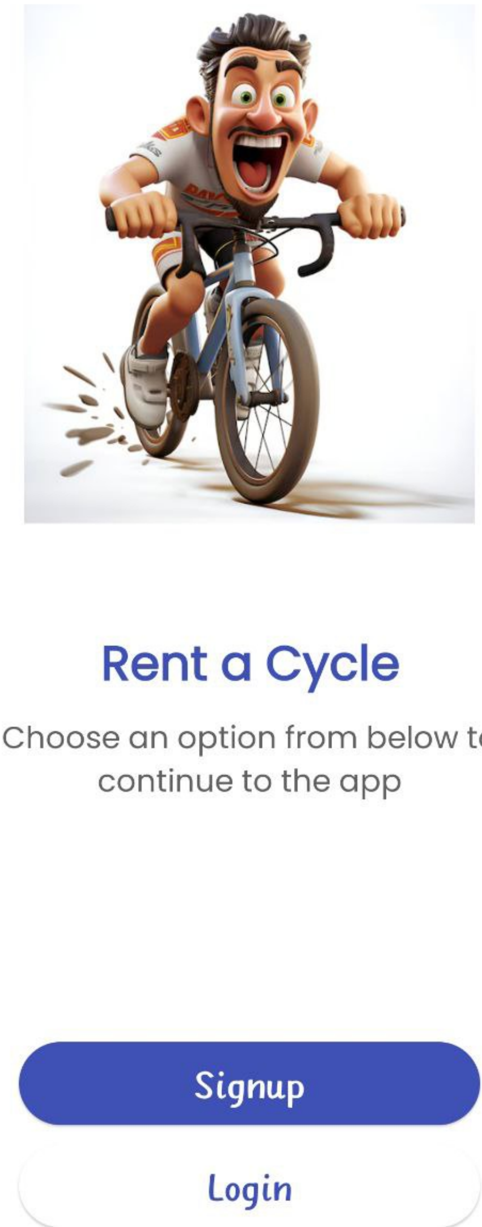

Figure 4.1: output 1

for users to check the bike's usage history, rental duration, and return location.

2. **Smart Lock System (ESP32 with Solenoid Lock):**

    - ESP32 Bluetooth Control: The ESP32 microcontroller acts as a bridge to control the solenoid lock, enabling secure bike lock/unlock functionality via Bluetooth.

- Bluetooth Connectivity: When users rent bikes, the system will use Bluetooth to unlock the bike automatically when the user is near the lock, allowing easy access. The lock will remain secured when the bike is not in use, preventing unauthorized access.

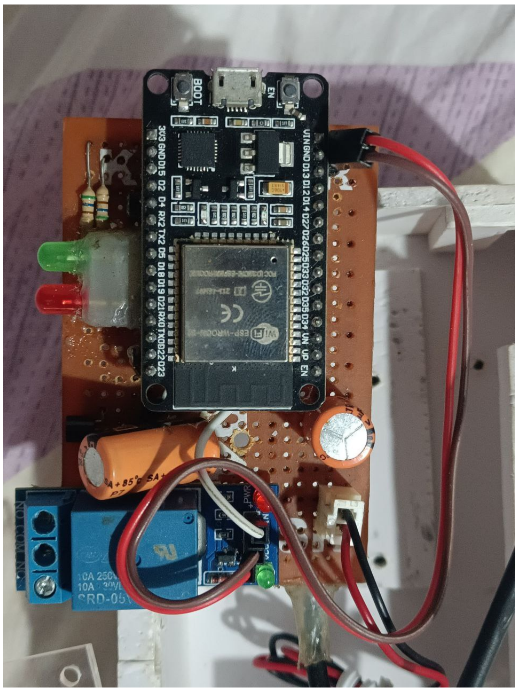

Figure 4.2: Hardware

- Smart Lock Features: Additional features like remote unlocking via the app, real-time status checking, or even automatic locking once the bike is returned might be included.

3. **Integrated System Workflow:**

- Admin Control: Admins would have control over the system through the app or a separate interface, allowing them to track bike availability, monitor usage, and manage payments.

4. **Security Features:**

- Secure Login: User login and signup processes would be secure, preventing unauthorized access.
- Bluetooth Security: The Bluetooth communication between the app and the bike lock would be encrypted, ensuring that the lock can only be controlled by the authorized user who rented the bike.
- Payment Security: Payments made through the app would be secure, following industry standards for payment gateways.

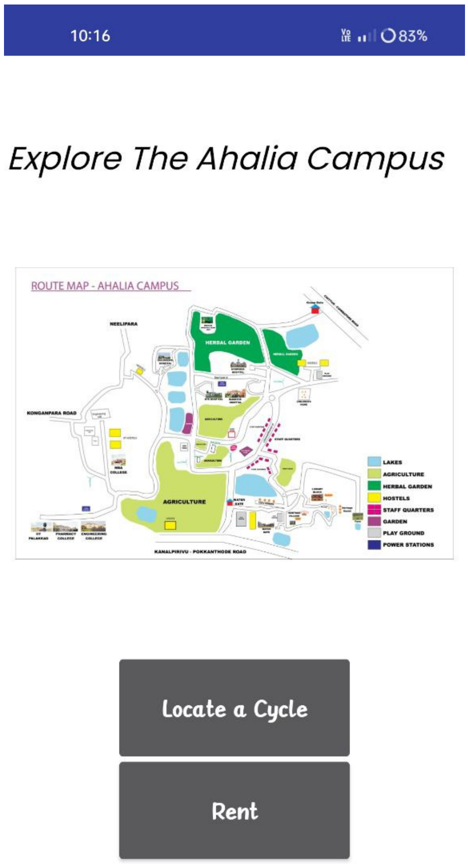

Figure 4.3: output 2

Overall the app developed for the bicycle rental system successfully combines functionality with an intuitive design. By offering a seamless selection process, informative content, and navigational tools, this project enhances the overall experience for users looking to explore the campus on two wheels.

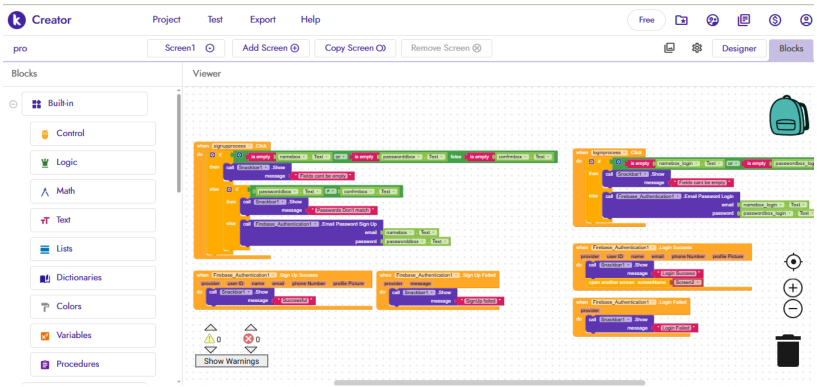

Figure 4.4: Code blocks

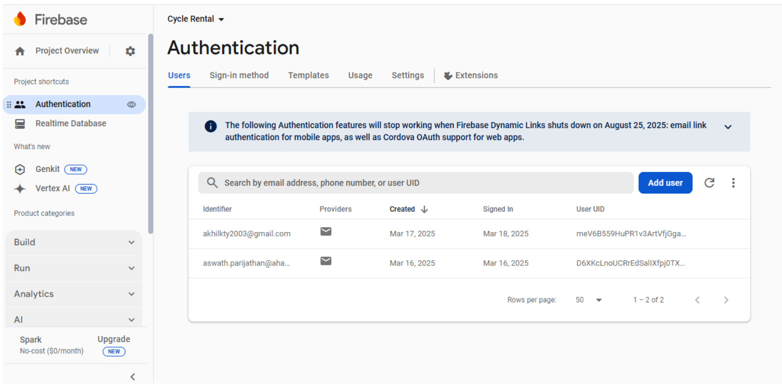

Figure 4.5: Database

Chapter 5

CONCLUSION

The smart bicycle sharing service offers a transformative solution for campus mobility by integrating technology with sustainability. Utilizing a mobile application, the service provides users with convenient access to bicycles, enabling easy reservations, real-time tracking, and secure payments. This initiative not only enhances individual mobility but also supports the university's sustainability goals by reducing reliance on motor vehicles and promoting healthier lifestyles. The system encourages a cycling culture on campus, making biking an appealing transportation option. For long-term success, ongoing user engagement, effective maintenance, and community partnerships are essential. This project stands as a model for innovative campus transportation, paving the way for a more sustainable and connected environment while setting a standard for future mobility solutions.

Here are some key areas for future development:

1. **Android application development**

    - **Mobile Access:** Creating a dedicated application for Android and IOS to allow users to rent bicycles on the go, providing a more convenient and accessible platform.

2. **GPS Tracking**

    - **Route Optimization:** Providing users with suggested cycling routes based on real-time traffic and path conditions, improving their overall experience.

- **Geofencing:** Setting up virtual boundaries to alert users if they take their bike outside designated areas, ensuring compliance with rental policies.

3. **Payment Deduction And Integration**

   - **Automatic Payment Deduction:** Implementing a system for automatic payment deductions based on the rental duration, making the process seamless for users.
   - **Promotional Offers and Discounts:** Allowing users to apply promo codes or discounts directly through the app, encouraging usage and customer loyalty

4. **User Account Management:**

   - **Profile Management:** Enabling users to create and manage their profiles, including rental history, preferences, and payment methods.

5. **Enhanced Analytics:**

   - **Data Collection:** Utilizing data analytics to gather insights on user behavior, rental patterns, and bike usage, which can inform future business strategies.

6. **Community Features:**

   - **Social Sharing:** Allowing users to share their biking experiences on social media directly from the app, promoting community engagement and brand awareness.

7. **Sustainability Initiatives:**

   - **Eco-Friendly Features:**Highlighting the environmental benefits of cycling and promoting sustainability initiatives, such as carbon offset programs for users.

Chapter 6

REFERENCES

1. Enhancement of Mobile-Based Application for Vehicle Rental Falah YH Ahmed, Eizwan Bin Hazlan, Muhammad Irsyad Abdulla 2021 IEEE 11th IEEE Symposium on Computer Applications Industrial Electronics (ISCAIE), 163-168, 2021

2. E. Fishman, S. Washington, and N. Haworth, ”Bike Share: A Synthesis of the Evidence,” Transportation Research Part A: Policy and Practice, vol. 34, no. 1, pp. 2-20, 2021.DOI: 10.1080/01441647.2013.848175.

3. T. M. H. Theesfeld, ”Exploring the Impact of Bicycle Sharing Systems on Urban Mobility,” Journal of Urban Planning and Development, vol. 146, no. 2, pp. 04020005, June 2020.DOI: 10.1061/(ASCE)UP.1943-5444.0000560.

4. D. M. Dill and T. W. Carr, ”Bicycle Commuting and Facilities in Portland, Oregon,” Transportation Research Record, vol. 2672, no. 2, pp. 63-70, 2021. DOI: 10.1177/03611981211008794.

5. Rent-A-Cycle (Smart Bicycle Sharing Service-IOT Based) Sarmad Hameed,Faraz Junejo, Naqi Jafri, Dania Rashid, Fabiha Shoaib J Robot Mech Eng 1, 1-10,2021

6. Teo and Wang, ”Design and Implementation of an Efficient Bicycle Rental System,” in Proceedings of the 2018 IEEE International Conference on Industrial Engineering and Engineering Management (IEEM), Bangkok, Thailand, 2020, pp. 222-226.DOI: 10.1109/IEEM.2018.8607472.

7. J. W. Wu, ”Smart Bicycle Rental System Based on IoT and Blockchain Technology,” IEEE Internet of Things Journal, vol. 7, no. 10, pp. 9802-9810, Oct. 2020. DOI: 10.1109/JIOT.2020.2995647

8. A. R. Becker, ”Economic Viability of Bike Sharing Programs: A Case Study,” IEEE Transactions on Transportation Electrification, vol. 7, no. 4, pp. 2076-2085, Dec. 2021. DOI: 10.1109/TTE.2021.3083294.

9. GlideX: An IoT Based Bicycle Rental System Lakshmi Boppana, Seth Ranjan Chodagam, Levin Joji Mathews, Vishnu Murali Krishnan, Indranil Abhay Athawle 2024 1st International Conference on Robotics, Engineering,Science, and Technology (RESTCON), 165-169, 2024

10. Crowdsourced Bicycle Monitoring and Finding with Riding User Recognition Based on Internet of Things Technologies Lien-Wu Chen, Chun-Yu Cho, Ai-Ni Li, Ming-Fong Tsai IEEE Sensors Journal, 2024

11. L. A. Moncayo-Martínez and A. Ramirez-Nafarrate, ”Visualization of the mobility patterns in the bike-sharing transport systems in Mexico City,” 2020 IEEE International Conference on Industrial Engineering and Engineering Management (IEEM), Bali, 2020, pp. 1851-1855, doi: 10.1109/IEEM.2020.7798198

12 J. Hu, J. Li and S. A. Nawaz, ”The design of a reminder device for placing shared bikes neatly based on attitude sensors,” 2021 IEEE International Conference on Advanced Manufacturing (ICAM), Yunlin, 2021, pp. 166-169, doi: 10.1109/AMCON.2021.8614774

IRJMETS

International Research Journal Of Modernization in Engineering Technology and Science

(Peer-Reviewed, Open Access, Fully Refereed International Journal)

e-ISSN: 2582-5208

Ref: IRJMETS/Certificate/Volume 07/Issue 03/70300115725

DOI: https://www.doi.org/10.56726/IRJMETS69556

Date: 29/03/2025

Certificate of Publication

*This is to certify that author "**Aswath P**" with paper ID "**IRJMETS70300115725**" has published a paper entitled "CAMPUS YIELDING CONVENIENT LOCKABLE ECO-BIKES (CYCLE)" **in International Research Journal Of Modernization In Engineering Technology And Science (IRJMETS), Volume 07, Issue 03, March 2025***

Editor in Chief

IRJMETS
Impact Factor
8.187

We Wish For Your Better Future

www.irjmets.com

Plagiarism Checker X - Report

Originality Assessment

4%

Overall Similarity

Date: Mar 31, 2025
Matches: 52 / 1487 words
Sources: 5

Remarks: Low similarity detected, consider making necessary changes if needed.

Verify Report:
Scan this QR Code

Sources

1	https://www.semanticscholar.org/author/Seth-Ranjan-Chodagam/2292320095 INTERNET 1%
2	https://sci-hub.se/10.1109/ISCAIE51753.2021.9431820 INTERNET 1%
3	https://link.springer.com/chapter/10.1007/978-3-031-66943-9_5 INTERNET <1%
4	https://tecnobits.com/en/introduction-to-google-maps-go-features-and-functionalities/ INTERNET <1%
5	https://quizlet.com/712131495/opera-cloud-accountingfinance-assessment-flash-cards/ INTERNET <1%

EXCLUDE CUSTOM MATCHES	ON
EXCLUDE QUOTES	OFF
EXCLUDE BIBLIOGRAPHY	OFF